Conditional Probability and Bayes' Theorem

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Outline

Conditional Probability

Conditional Probability

Definition

Conditional probability is the likelihood of an event occurring, assuming that another event has **already occurred**.

Notation

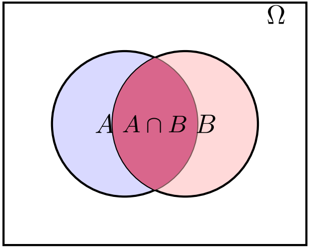
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Read as: "Probability of A **given** B."

Visualizing Conditional Probability

The Intuition:

- The original sample space is everything (the white box).
- Once we know B has happened, our "world" shrinks.
- The new sample space is just the circle B.
- We look for the portion of A that is **inside** B.



Why Learn This? ML Applications

Conditional probability is the engine behind many ML algorithms.

1. **Classification (Naive Bayes):**
   - Given an email contains the word “Free”, what is the probability it is Spam?
   - $P(\text{Spam} \mid \text{“Free”})$
2. **Natural Language Processing (NLP):**
   - Next Word Prediction (LLMs like GPT).
   - Given the previous word is “Artificial”, what is the prob the next word is “Intelligence”?
   - $P(\text{“Intelligence”} \mid \text{“Artificial”})$
3. **Recommendation Systems:**
   - Given a user bought a Laptop, what is the probability they buy a Mouse?
   - $P(\text{Mouse} \mid \text{Laptop})$

Conditional Probability Ex 1: The “Tech Company” Scenario

Scenario: In a company of 100 employees:

- 40 are Data Scientists (DS).
- 25 employees use Python.
- 15 are Data Scientists AND use Python.

Question: If we pick a random employee and see they are a Data Scientist, what is the probability they use Python?

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Question: If we pick a random employee and see they are a Data Scientist, what is the probability they use Python?

Solution:

1. Event A: Uses Python. Event B: Is a Data Scientist.
2. $P(B) = 40/100 = 0.4$.
3. $P(A \cap B) = 15/100 = 0.15$.
4. Formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.40} = \mathbf{0.375}$$

Conditional Probability Ex 2: Rain and Traffic

Scenario:

- Probability of Rain (R) = 20%.
- Probability of Heavy Traffic (T) = 35%.
- Probability of Rain AND Heavy Traffic = 15%.

Question: You look out the window and see it is raining. What is the probability you will encounter heavy traffic?

Conditional Probability Ex 2: Rain and Traffic

Scenario:

- Probability of Rain $(R) = 20\%$.
- Probability of Heavy Traffic $(T) = 35\%$.
- Probability of Rain AND Heavy Traffic $= 15\%$.

Question: You look out the window and see it is raining. What is the probability you will encounter heavy traffic?

Solution:

$$P(T \mid R) = \frac{P(T \cap R)}{P(R)}$$

$$P(T \mid R) = \frac{0.15}{0.20} = \mathbf{0.75}$$

Interpretation: Rain significantly increases the chance of traffic (from 35% to 75%).

Conditional Probability Ex 3: Online Shopping

Scenario: An e-commerce site analyzes customer behavior.

- 60% of visitors add an item to the cart (C).
- 20% of visitors proceed to checkout (O).
- (Note: You must add to cart to checkout, so O implies C, meaning $P(O \cap C) = P(O) = 0.20$).

Question: Given a customer has added an item to the cart, what is the probability they will checkout?

Conditional Probability Ex 3: Online Shopping

Scenario: An e-commerce site analyzes customer behavior.

- 60% of visitors add an item to the cart (C).
- 20% of visitors proceed to checkout (O).
- (Note: You must add to cart to checkout, so O implies C, meaning $P(O \cap C) = P(O) = 0.20$).

Question: Given a customer has added an item to the cart, what is the probability they will checkout?

Solution:

$$P(O \mid C) = \frac{P(O \cap C)}{P(C)} = \frac{0.20}{0.60} = \frac{1}{3} \approx \mathbf{0.33}$$

Interpretation: The "Cart Abandonment Rate" is $1 - 0.33 = 0.67$ *or 67%.*

Multiplication Rule

We can rewrite the conditional probability formula as:

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

Generalizing to n Events (The Chain Rule): For events $A_1, A_2, \ldots, A_n$:

$$P(A_1 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot \ldots P(A_n \mid A_1 \ldots A_{n-1})$$

This is fundamental in sequential decision making (like Reinforcement Learning).

Multiplication Rule Ex 1: Drawing Cards

Problem: You draw 3 cards from a deck of 52 without replacement. What is the probability they are all Aces?

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Solution: Let A_i be the event the i-th card is an Ace.

$$\begin{aligned} P(A_1 \cap A_2 \cap A_3) &= P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \\ &= \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} \\ &= \frac{1}{13} \times \frac{1}{17} \times \frac{1}{25} \\ &\approx \mathbf{0.00018} \end{aligned}$$

Multiplication Rule Ex 2: The Job Interview

Problem: A candidate must pass 3 rounds of interviews to get hired.

- Round 1 Pass Rate: 50% ($P(R_1)$).
- Round 2 Pass Rate (given passed R1): 40% ($P(R_2|R_1)$).
- Round 3 Pass Rate (given passed R1 & R2): 30% ($P(R_3|R_1, R_2)$).

What is the probability of getting hired?

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- Round 3 Pass Rate (given passed R1 & R2): 30% ($P(R_3|R_1, R_2)$).

What is the probability of getting hired?

Solution:

$$P(\text{Hired}) = 0.5 \times 0.4 \times 0.3 = \mathbf{0.06}$$

Interpretation: Only a 6% chance of navigating the entire pipeline.

Independent Events

Two events are **Independent** if knowing one occurred gives us **no information** about the other.

Definition

Events A and B are independent if:

$$P(A \mid B) = P(A)$$

Or equivalently:

$$P(A \cap B) = P(A) \cdot P(B)$$

Independence vs. Disjoint (Mutually Exclusive)

Confusion often arises here. Let's distinguish them using a Card Deck.

Disjoint (Mutually Exclusive)

- **Example:** Event A (Card is a King), Event B (Card is a Queen).
- Can they happen at the same time? **No.**
- $P(A \cap B) = 0$.
- If I see a King, I know for a fact it is **not** a Queen. (Highly Dependent!)

Independent

- **Example:** Event A (Card is a King), Event B (Card is Red).
- Can they happen at the same time? **Yes** (Red King).
- $P(A) = 4/52$.
  $P(A \mid Red) = 2/26 = 4/52$.
- Knowing the card is Red doesn't change the odds of it being a King.

Numerical Example: Disjoint vs Independent

Experiment: Rolling a fair 6-sided die.

Case 1: Disjoint Events

- $A = \{1, 2\}$, $B = \{5, 6\}$.
- $P(A) = 1/3$, $P(B) = 1/3$.
- $P(A \cap B) = 0$.
- Check Independence: Does $P(A \cap B) = P(A)P(B)$?
- $0 \neq 1/9$. **Not Independent.**

Case 2: Independent Events

- $A = \{1, 2\}$ (Low number), $C = \{2, 4, 6\}$ (Even number).
- $P(A) = 2/6 = 1/3$. $P(C) = 3/6 = 1/2$.
- Intersection: $A \cap C = \{2\}$. So $P(A \cap C) = 1/6$.
- Check Independence: Does $P(A \cap C) = P(A)P(C)$?
- $1/6 = (1/3) \times (1/2)$. **Yes, Independent.**

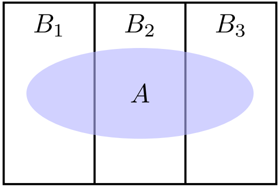
Total Probability and Bayes' Theorem

Theorem of Total Probability

If events $B_1, B_2, \ldots, B_n$ partition the sample space (they are disjoint and cover everything), then for any event A:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) \cdot P(B_i)$$

Why? We break a complex event A into smaller, manageable pieces based on the conditions B_i.



Total Probability Ex 1: Stock Market

Scenario: A stock's performance depends on the general economy state.

- Economy States: Bull (40%), Bear (30%), Stagnant (30%).
- Prob of Stock Rising in Bull market: 80%.
- Prob of Stock Rising in Bear market: 20%.
- Prob of Stock Rising in Stagnant market: 50%.

Q: What is the overall probability the stock rises?

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- Prob of Stock Rising in Bear market: 20%.
- Prob of Stock Rising in Stagnant market: 50%.

Q: What is the overall probability the stock rises?

Solution:

$$\begin{aligned} P(Rise) &= P(R \mid Bull)P(Bull) + P(R \mid Bear)P(Bear) + P(R \mid Stag)P(Stag) \\ &= (0.8 \times 0.4) + (0.2 \times 0.3) + (0.5 \times 0.3) \\ &= 0.32 + 0.06 + 0.15 \\ &= \mathbf{0.53} \end{aligned}$$

Total Probability Ex 2: Email Servers

Scenario: Your company receives emails through 2 servers.

- Server A handles 70% of traffic; Server B handles 30%.
- Server A blocks an email as spam erroneously 1% of the time.
- Server B blocks an email as spam erroneously 2.5% of the time.

Q: What is the total probability a legitimate email gets blocked?

Total Probability Ex 2: Email Servers

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- Server B blocks an email as spam erroneously 2.5% of the time.

Q: What is the total probability a legitimate email gets blocked?

Solution:

$$\begin{aligned} P(Block) &= P(B \mid S_A)P(S_A) + P(B \mid S_B)P(S_B) \\ &= (0.01 \times 0.70) + (0.025 \times 0.30) \\ &= 0.007 + 0.0075 \\ &= \mathbf{0.0145} \quad (1.45\%) \end{aligned}$$

Total Probability Ex 3: Factory Production

Scenario: A factory has 3 machines making chips.

- Machine 1: Makes 50% of chips, 1% defect rate.
- Machine 2: Makes 30% of chips, 2% defect rate.
- Machine 3: Makes 20% of chips, 3% defect rate.

Q: What is the total probability of picking a defective chip?

Total Probability Ex 3: Factory Production

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- Machine 1: Makes 50% of chips, 1% defect rate.
- Machine 2: Makes 30% of chips, 2% defect rate.
- Machine 3: Makes 20% of chips, 3% defect rate.

Q: What is the total probability of picking a defective chip?

Solution: Let D = Defective.

$$\begin{aligned} P(D) &= P(D \mid M_1)P(M_1) + P(D \mid M_2)P(M_2) + P(D \mid M_3)P(M_3) \\ &= (0.01)(0.50) + (0.02)(0.30) + (0.03)(0.20) \\ &= 0.005 + 0.006 + 0.006 \\ &= 0.017 \end{aligned}$$

Total defect rate is **1.7%**.

Bayes' Theorem

Bayes' Theorem allows us to "flip" conditional probabilities. It updates our belief about an event based on new evidence.

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Expanded using Total Probability:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)}$$

$\neg A$ is same as A^c, which means complementary event.

Terminologies

- $P(A)$: **Prior** (What we thought before evidence / data).
- $P(B \mid A)$: **Likelihood** (How probable is the evidence / data given our hypothesis?).
- $P(B)$: **Evidence** (Total probability of the evidence / data).
- $P(A \mid B)$: **Posterior** (Updated belief after seeing data / evidence).

Bayes' Theorem Ex 1: Employee Retention

Scenario:

- Probability an employee leaves the company this year ($P(L)$): 10%.
- If an employee is leaving, there is a 60% chance they logged in remotely late at night ($P(N \mid L) = 0.6$).
- If an employee is STAYING, there is only a 10% chance they logged in late at night ($P(N \mid \neg L) = 0.1$).

Q: We observe a user logging in late at night. What is the prob they are leaving?

Bayes' Theorem Ex 1: Employee Retention

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- Probability an employee leaves the company this year ($P(L)$): 10%.
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- If an employee is STAYING, there is only a 10% chance they logged in late at night ($P(N \mid \neg L) = 0.1$).

Q: We observe a user logging in late at night. What is the prob they are leaving?

Solution:

$$P(L|N) = \frac{0.6 \times 0.1}{(0.6 \times 0.1) + (0.1 \times 0.9)} = \frac{0.06}{0.06 + 0.09} = \frac{0.06}{0.15} = \mathbf{0.40}$$

The probability jumps from 10% to 40%.

Bayes' Theorem Ex 2: Defective Production

Scenario: Machine A produces 30% of parts, Machine B produces 70%.

- Machine A defect rate: 5%.
- Machine B defect rate: 1%.

Q: A part is found to be defective. What is the probability it came from Machine A?

Bayes' Theorem Ex 2: Defective Production

Scenario: Machine A produces 30% of parts, Machine B produces 70%.

- Machine A defect rate: 5%.
- Machine B defect rate: 1%.

Q: A part is found to be defective. What is the probability it came from Machine A?

Solution: Let D = Defective.

$$\begin{aligned} P(A \mid D) &= \frac{P(D \mid A)P(A)}{P(D \mid A)P(A) + P(D \mid B)P(B)} \\ &= \frac{0.05 \times 0.30}{(0.05 \times 0.30) + (0.01 \times 0.70)} \\ &= \frac{0.015}{0.015 + 0.007} = \frac{0.015}{0.022} \approx \mathbf{0.68} \end{aligned}$$

Even though Machine A makes fewer parts, its high defect rate makes it the likely culprit (68%).

Bayes' Theorem Ex 3: The Spam Filter

Context:

- 40% of all email is Spam (S).
- 60% is Ham (Not Spam) (H).
- The word "Offer" appears in 25% of Spam emails.
- The word "Offer" appears in only 5% of Ham emails.

Q: An email arrives containing the word "Offer". What is the probability it is Spam?

Bayes' Theorem Ex 3: The Spam Filter

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- 60% is Ham (Not Spam) (H).
- The word "Offer" appears in 25% of Spam emails.
- The word "Offer" appears in only 5% of Ham emails.

Q: An email arrives containing the word "Offer". What is the probability it is Spam?

Solution:

$$
\begin{aligned}
P(S \mid \text{“Offer”}) &= \frac{P(\text{“Offer”} \mid S)P(S)}{P(\text{“Offer”} \mid S)P(S) + P(\text{“Offer”} \mid H)P(H)} \\
&= \frac{0.25 \times 0.40}{(0.25 \times 0.40) + (0.05 \times 0.60)} \\
&= \frac{0.10}{0.10 + 0.03} \\
&= \frac{0.10}{0.13} \approx \mathbf{0.77}
\end{aligned}
$$

Bayes' Theorem Ex 4: Diagnostic Test

Context: A disease affects 0.1% of the population ($P(D) = 0.001$).

- Test True Positive Rate: 99% ($P(+ \mid D) = 0.99$).
- Test False Positive Rate: 2% ($P(+ \mid \neg D) = 0.02$).

Q: Patient tests positive. Probability they have the disease?

Bayes' Theorem Ex 4: Diagnostic Test

Context: A disease affects 0.1% of the population ($P(D) = 0.001$).

- Test True Positive Rate: 99% ($P(+ \mid D) = 0.99$).
- Test False Positive Rate: 2% ($P(+ \mid \neg D) = 0.02$).

Q: Patient tests positive. Probability they have the disease?

$$
\begin{aligned}
P(D) = 0.001, &\quad P(\neg D) = 0.999 \\
P(+ \mid D) = 0.99, &\quad P(+ \mid \neg D) = 0.02 \\
P(D \mid +) &= \frac{0.99 \times 0.001}{(0.99 \times 0.001) + (0.02 \times 0.999)} \\
&= \frac{0.00099}{0.00099 + 0.01998} \\
&= \frac{0.00099}{0.02097} \approx \mathbf{0.047} \quad (\approx 4.7\%)
\end{aligned}
$$

Result: *Even with a positive test, there is only a $\approx 5\%$ chance the person has the disease!*

Test your skills

Quiz Q1: Independent or Not?

Problem: You roll a red die and a blue die.

- Event A: The sum of the dice is 7.
- Event B: The red die shows a 4.

Are A and B independent? Calculate to prove.

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- Event A: The sum of the dice is 7.
- Event B: The red die shows a 4.

Are A and B independent? Calculate to prove.

Answer:

- $P(B) = 1/6$.
- Combinations for Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Total $6/36 = 1/6$. So $P(A) = 1/6$.
- Intersection (Sum is 7 AND Red is 4): Only one case (Red 4, Blue 3). $P(A \cap B) = 1/36$.
- Check: $(1/6) \times (1/6) = 1/36$.
- **Yes, they are Independent!** Knowing the Red is 4 does not change the probability of the sum being 7.

Quiz Q2: The “Fraud” Alert

Problem: A credit card fraud system detects 100% of real frauds, but also flags 1% of legitimate transactions as fraud (false alarm). Real fraud is rare: 0.1% of all transactions.
If the system flags a transaction, what is the probability it is actually fraud?

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If the system flags a transaction, what is the probability it is actually fraud?

Answer:

$$\frac{1.0 \times 0.001}{(1.0 \times 0.001) + (0.01 \times 0.999)} = \frac{0.001}{0.001 + 0.00999} \approx \frac{0.001}{0.011} \approx \mathbf{9\%}$$

Most flags are false alarms!

Quiz Q3: The “Twin Detectives” (Combined Probability)

Problem: Two detectives, Alice and Bob, are working on a case independently.

- Probability Alice solves it: 70%.
- Probability Bob solves it: 60%.

What is the probability that the case gets solved (by at least one of them)?

Quiz Q3: The "Twin Detectives" (Combined Probability)

Problem: Two detectives, Alice and Bob, are working on a case independently.

- Probability Alice solves it: 70%.
- Probability Bob solves it: 60%.

What is the probability that the case gets solved (by at least one of them)?

Answer: Easier to calculate probability that **Neither** solves it.

- $P(\text{Alice Fails}) = 0.3$.
- $P(\text{Bob Fails}) = 0.4$.
- $P(\text{Both Fail}) = 0.3 \times 0.4 = 0.12$ (Assuming Independence).
- $P(\text{Solved}) = 1 - 0.12 = \mathbf{0.88}$ (88%).

Quiz Q4: Sequence Prediction

Problem: A user visits a website.

- If they are on the Home Page, 50% go to "Products".
- If they are on "Products", 30% go to "Buy".

What is the probability a random user lands on Home, goes to Products, and then Buys?

Quiz Q4: Sequence Prediction

Problem: A user visits a website.

- If they are on the Home Page, 50% go to "Products".
- If they are on "Products", 30% go to "Buy".

What is the probability a random user lands on Home, goes to Products, and then Buys?

Answer: This is a Chain Rule application.

$$P(H, P, B) = P(H) \times P(P \mid H) \times P(B \mid P)$$

Assuming everyone starts at Home ($P(H) = 1$):

$$1 \times 0.50 \times 0.30 = \mathbf{0.15} \text{ or } 15\%$$

Quiz Q5: The Urn Challenge

Problem: Urn A has 2 Red balls, 1 Blue. Urn B has 1 Red, 2 Blue. You flip a fair coin. Heads $\rightarrow$ Pick from Urn A. Tails $\rightarrow$ Pick from Urn B.
You picked a **Red** ball. What is the probability you flipped Heads (picked from Urn A)?

Quiz Q5: The Urn Challenge

Problem: Urn A has 2 Red balls, 1 Blue. Urn B has 1 Red, 2 Blue. You flip a fair coin. Heads $\rightarrow$ Pick from Urn A. Tails $\rightarrow$ Pick from Urn B.
You picked a **Red** ball. What is the probability you flipped Heads (picked from Urn A)?

Answer (Bayes):

- Prior: $P(Heads) = 0.5$.
- Likelihoods: $P(Red \mid Heads) = 2/3$. $P(Red \mid Tails) = 1/3$.

$$P(H|R) = \frac{(2/3 \times 1/2)}{(2/3 \times 1/2) + (1/3 \times 1/2)} = \frac{2/6}{2/6 + 1/6} = \frac{2}{3} \approx \mathbf{0.66}$$

Summary & Next Steps

Summary

- **Conditional Probability:** Updating probability based on new info.
- **Independence:** Crucial assumption for simplifying ML models.
- **Bayes' Theorem:** The mathematical foundation for inference.

Next Steps

Next we shall learn about Probability Distribution (Discrete and Continuous)

Thank You!